CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

CDK4/6: 4 molecules stopped early, all small molecules

6 stopped trials, but 4 molecules — all small molecules: Abemaciclib, Palbociclib, CDK4/6 Inhibitor SPH4336 and Ribociclib. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

4.1%
6 of 147 closed trials (95% CI 1.9%–8.6%)

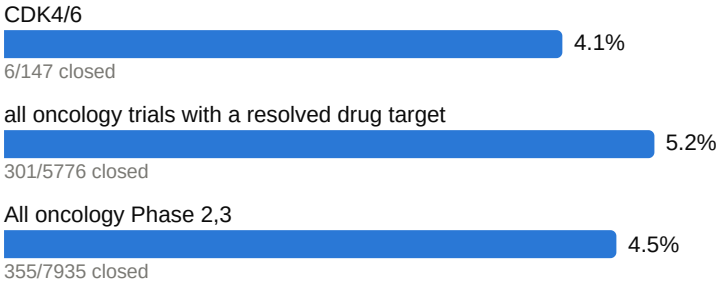
5 / 1 / 0
efficacy / safety / both (adds to 6)

39.8%
of 369 trials have closed · 222 still open or unresolved

1.4%
stopped for a biological reason within 3 years of starting (95% CI 0.5%–3.6%; 228 still at risk; 1.5% area-wide)

3 of 10 **Completed and missed the primary endpoint.** Of 102 completed trials in this segment, 10 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 3 missed and 7 met. Counted separately and never in the rate above; a floor, not a rate. 35 of the 99 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 242 distinct experimental drugs across 369 trials (small molecule 367, antibody 71, unknown 45, protein 19). Most tested: PALBOCICLIB (142), ABEMACICLIB (107), RIBOCICLIB (67), LETROZOLE (64), FULVESTRANT (57). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 6 stops came from 6 sponsor-asset programmes across 6 sponsors; the largest (Washington University School of Medicine) contributed 1; removing that programme's trials from both sides leaves 3.4%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 9 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 10.2%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	3	64	4.7%	1.6%–12.9%
2018–2020	2	54	3.7%	1.0%–12.5%
2021–2024	1	29	3.4%	0.6%–17.2%

Where the stops sit

Efficacy / futility: 5, Safety: 1.
Sponsors with most stops: Shanghai Pharma Biotherapeutics USA Inc. (1), Anita Turk (1), Sidney Kimmel Comprehensive Cancer Center at Thomas Jefferson University (1), Eli Lilly (1), Washington University School of Medicine (1), UNC Lineberger Comprehensive Cancer Center (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05580588	2	Shanghai Pharma Biotherapeutics USA Inc.	CDK4/6 Inhibitor SPH4336	2024-10	Efficacy	Although no safety issues with SPH4336 were identified, efficacy in liposarcoma patient was less than anticipated.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04534283	2	Anita Turk	ABEMACICLIB, TEMUTERKIB	2023-08	Efficacy	Lack of efficacy.
NCT02555189	2	Sidney Kimmel Comprehensive Cancer Center at Thomas Jefferson University	RIBOCICLIB	2022-12	Efficacy	Sponsor-PI decision. An analysis of the primary endpoint concluded that the addition of ribociclib to standard of care enzalutamide did not demonstrate benefit, an all active subjects were removed from the study.
NCT04165031	2	Eli Lilly LLY	LY-3499446, ABEMACICLIB, CETUXIMAB, ERLOTINIB	2020-10	Safety	The study was terminated due to an unexpected toxicity finding.
NCT02907918	2	Washington University School of Medicine	PALBOCICLIB, LETROZOLE, TRASTUZUMAB, GOSERELIN	2020-08	Efficacy	Futility
NCT02334527	2	UNC Lineberger Comprehensive Cancer Center	PALBOCICLIB	2016-12	Efficacy	Data from first 12 subjects-primary endpoint not met. Data analysis underway.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 39.8% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.